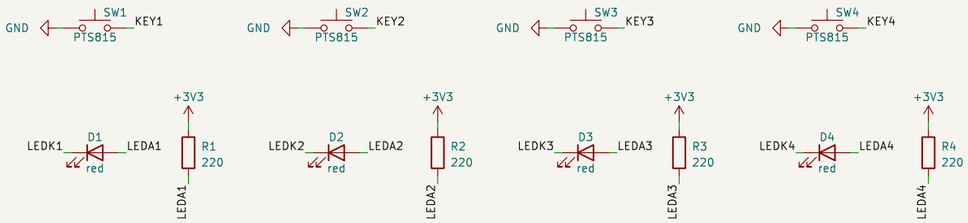
OpenTheatreComms beltpack — top PCB

Talk keys and key lights
C&K PTS815 under each keycap, red 0603 LED inside the cap, about 5 mA

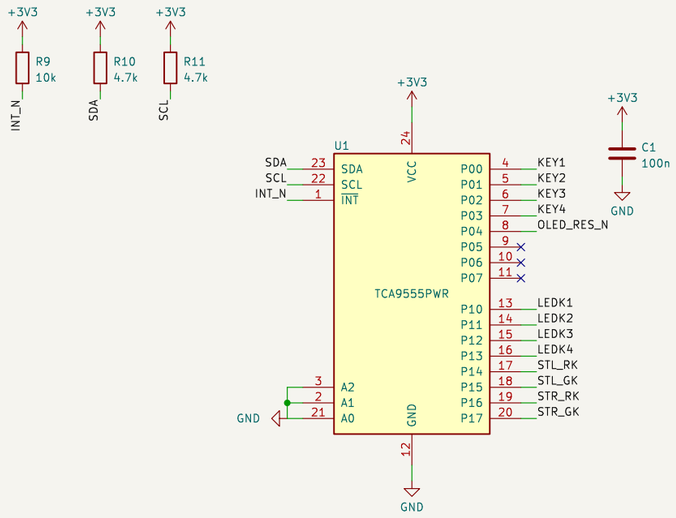


Status lights
Red + green 0603 pair under each end light pipe

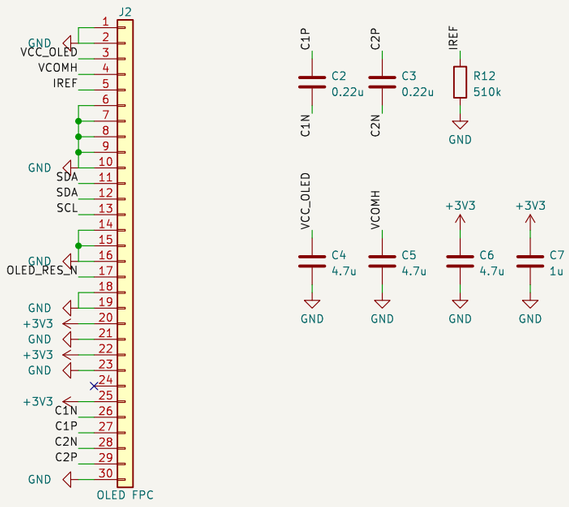


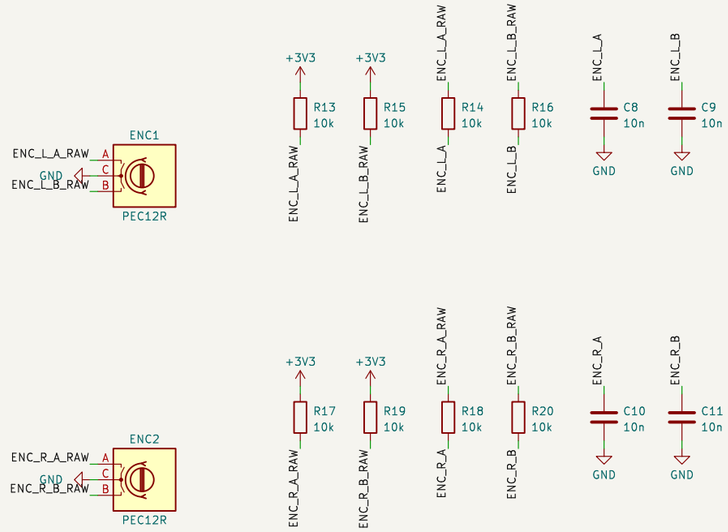
Dials: Bourns PEC12R-4215F-N0024 on the B side
Datasheet RC filter on A and B (10k pull-up, 10k series, 10 nF)

IO expander, I2C address 0x20
Keys in, key/status lights and OLED reset out.
The I2C pull-ups live here; leave them off the main PCB.

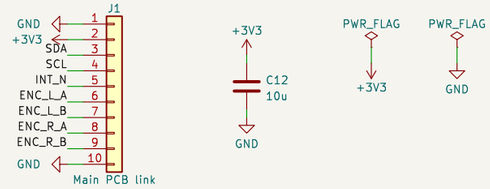


OLED: 1.3 in 128x64 white, GME12864-110 (SH1106)
I2C mode, address 0x3C. 30-pin 0.5 mm FPC into a ZIF on B.
Check the pin map and contact side against your panel's drawing.

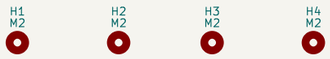




Link to the main PCB: JST SH 10-way, 50 mm cable
1 GND, 2 +3V3, 3 SDA, 4 SCL, 5 INT, 6–9 dial A/B, 10 GND



M2 mounting holes into the top-plate bosses



Keys, key and status lights, OLED, dials, IO expander. Generated from pcb_top/design.py

OpenTheatreComms

Sheet: /
File: top_pcb.kicad_sch

Title: Beltpack top PCB

Size: A3	Date: 2026-10-26	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1